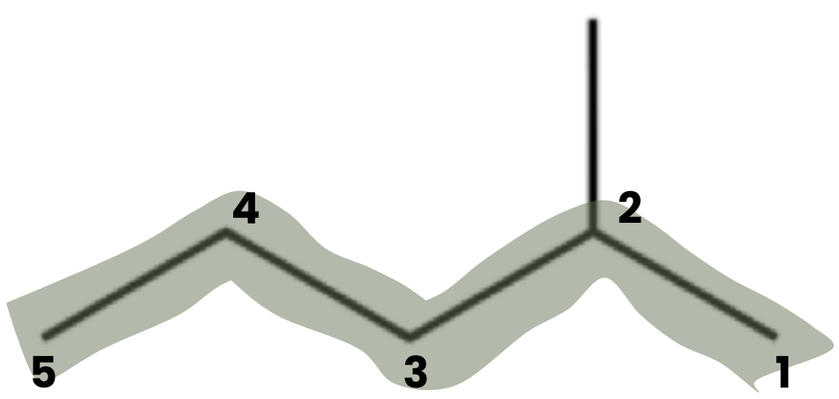
A01 FOUNDATION

C_6H_{14}

PART A
STRUCTURE
> NAME

1. NUMBERED PARENT



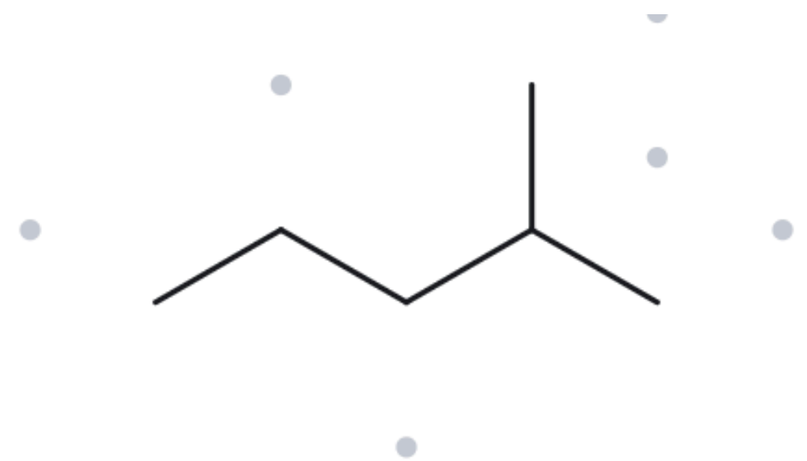
4. NAMING TRAIL

PARENT: LONGEST CHAIN = 5 C → PENTANE.
NUMBERING: THE METHYL IS ON C2 FROM THE END NEARER THE BRANCH AND ON C4 FROM THE OTHER END. FIRST POINT OF DIFFERENCE: 2 < 4, SO NUMBER FROM THE END NEARER THE BRANCH.
SUBSTITUENTS: METHYL ON C2 → 2-METHYL.

3. MY NAME

2-methylpentane

5. THE ALKANE CHECK



2-methylpentane

MATCH

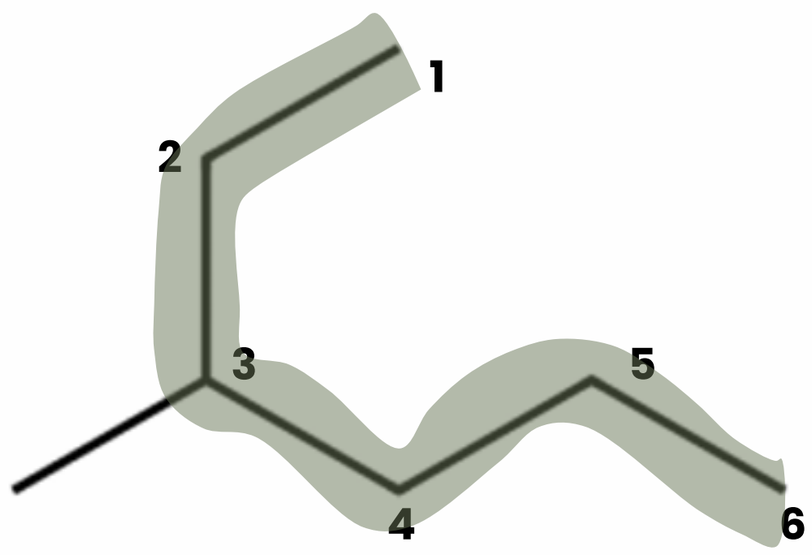
- ☑ LONGEST CHAIN CONFIRMED
- ☑ ALPHABETICAL ORDER
- ☑ FORMULA COUNTED
- ☑ LOWEST LOCANTS
- ☑ PUNCTUATION

A02 FOUNDATION

C_7H_{16}

PART A
STRUCTURE
> NAME

1. NUMBERED PARENT



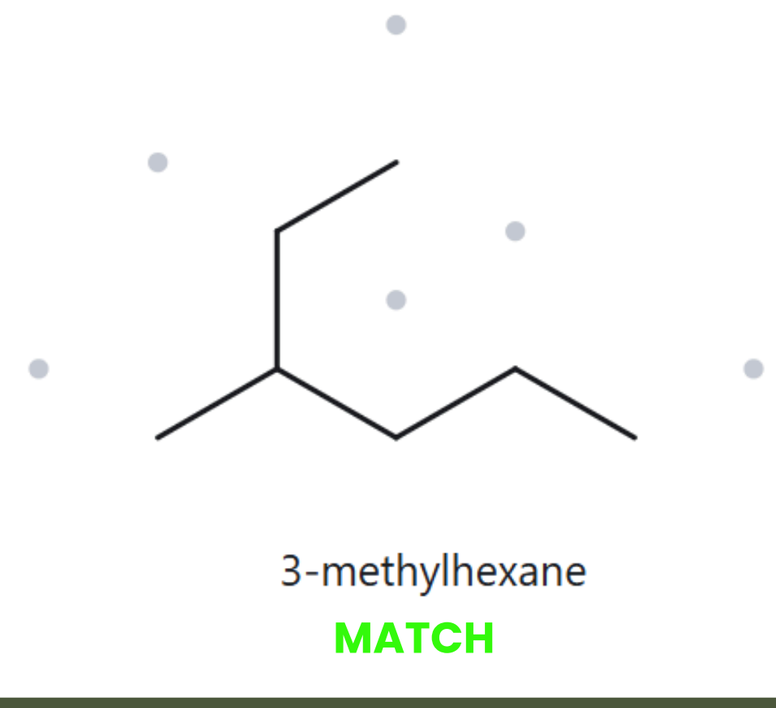
2. NAMING TRAIL

PARENT: LONGEST CHAIN = 6 C → HEXANE. THE LONGEST PATH RUNS THROUGH THE ETHYL BRANCH, NOT ALONG THE HORIZONTAL LINE.
NUMBERING: THE METHYL IS ON C3 FROM THE ETHYL END AND ON C4 FROM THE OTHER END. FIRST POINT OF DIFFERENCE: 3 < 4, SO NUMBER FROM THE ETHYL END.
SUBSTITUENTS: METHYL ON C3 → 3-METHYL.

3. MY NAME

3-methylhexane

4. NAMING TRAIL



5. THE ALKANE CHECK

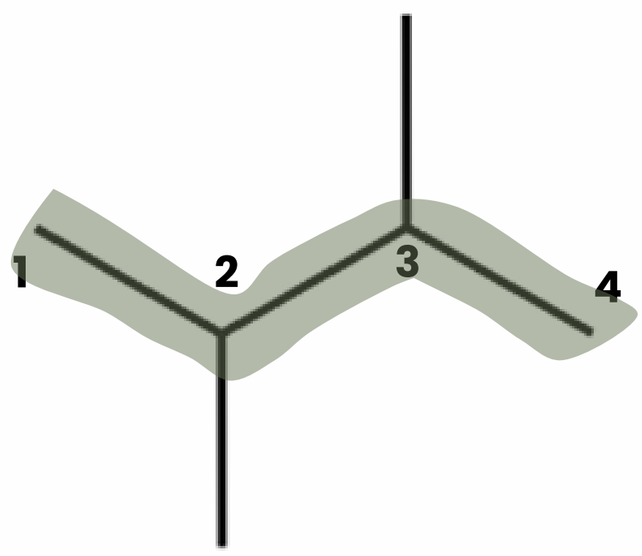
- ☑ LONGEST CHAIN CONFIRMED
- ☑ LOWEST LOCANTS
- ☑ ALPHABETICAL ORDER
- ☑ PUNCTUATION
- ☑ FORMULA COUNTED

A03 FOUNDATION

C_6H_{14}

PART A
STRUCTURE
> NAME

1. NUMBERED PARENT



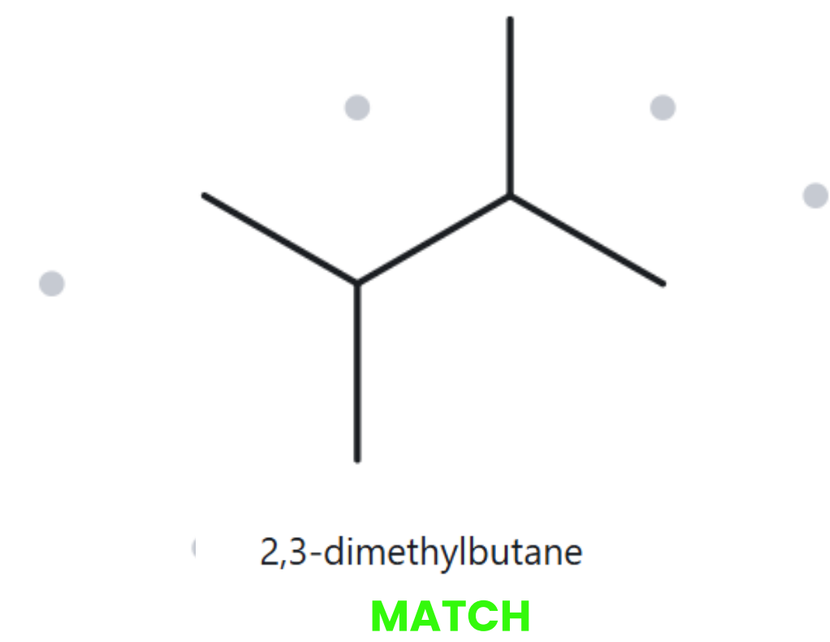
4. NAMING TRAIL

PARENT: LONGEST CHAIN = 4 C → BUTANE.
NUMBERING: BOTH DIRECTIONS GIVE 2,3, SO THE NUMBERING IS THE SAME EITHER WAY.
SUBSTITUENTS: METHYL ON C2 AND C3 → 2,3-DIMETHYL.

3. MY NAME

2,3-dimethylbutane

5. THE ALKANE CHECK



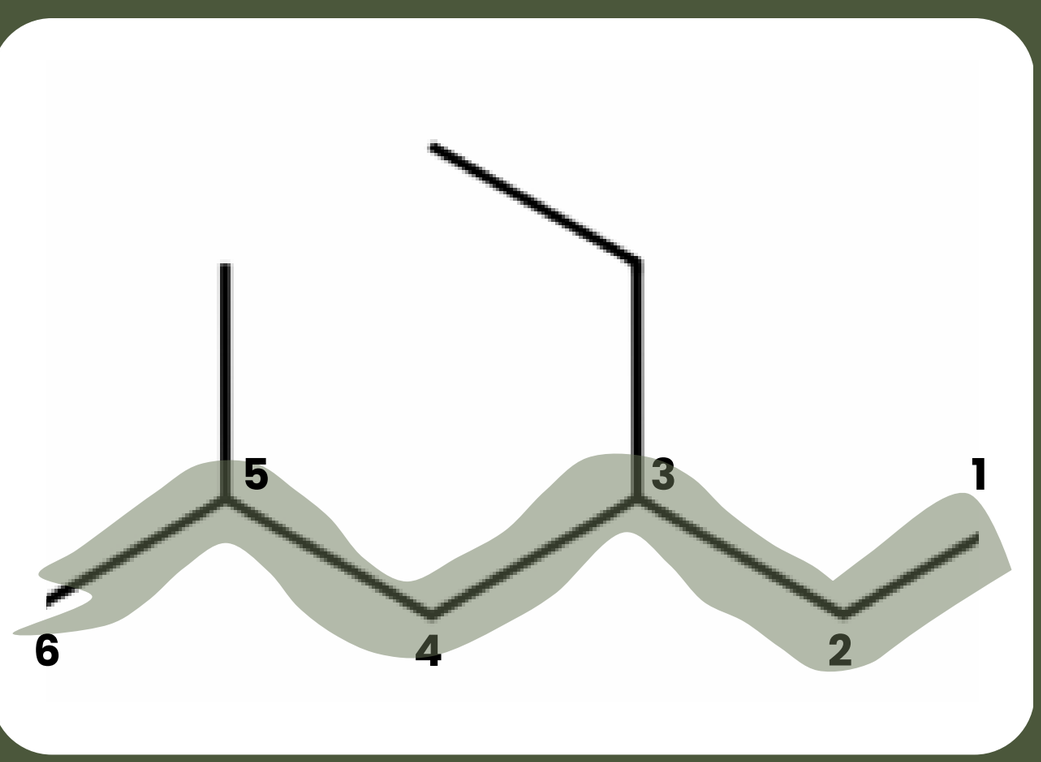
- ☑ LONGEST CHAIN CONFIRMED
- ☑ LOWEST LOCANTS
- ☑ ALPHABETICAL ORDER
- ☑ PUNCTUATION
- ☑ FORMULA COUNTED

A04 DEVELOPING

C_9H_{20}

PART A
STRUCTURE
> NAME

1. NUMBERED PARENT



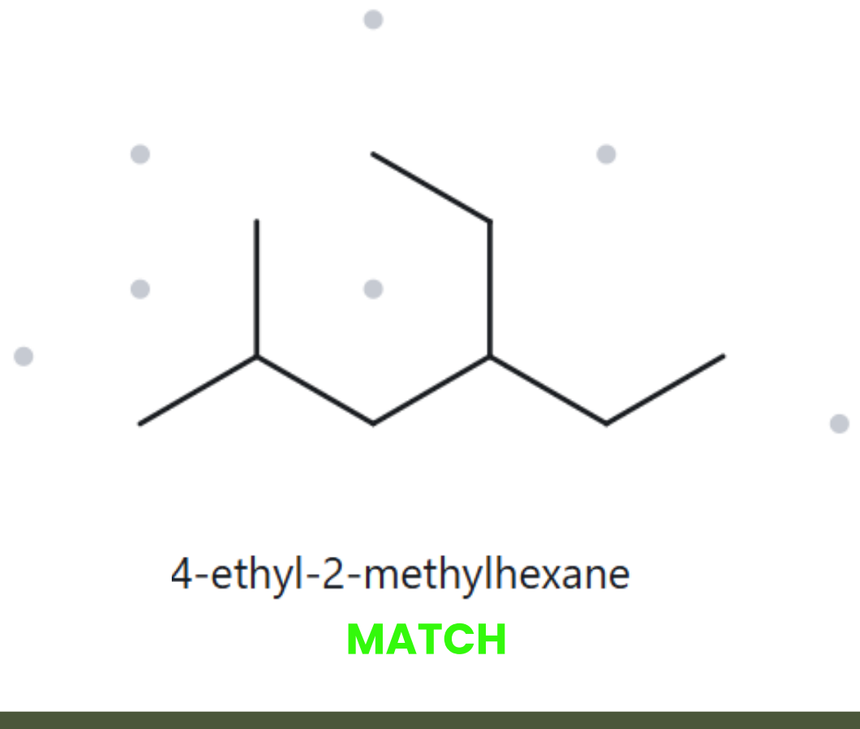
4. NAMING TRAIL

PARENT: LONGEST CHAIN = 6 C → HEXANE. SEVERAL 6-C PATHS EXIST, BUT EACH CARRIES ONE ETHYL AND ONE METHYL, SO THEY LEAD TO THE SAME NAME.
NUMBERING: LEFT → RIGHT GIVES 2,4; RIGHT → LEFT GIVES 3,5. FIRST POINT OF DIFFERENCE: 2 < 3, SO NUMBER FROM THE LEFT.
SUBSTITUENTS: ETHYL ON C4, METHYL ON C2. ALPHABETICAL ORDER PUTS ETHYL BEFORE METHYL → 4-ETHYL-2-METHYL.

3. MY NAME

4-ethyl-2-methylhexane

5. THE ALKANE CHECK



4-ethyl-2-methylhexane
MATCH

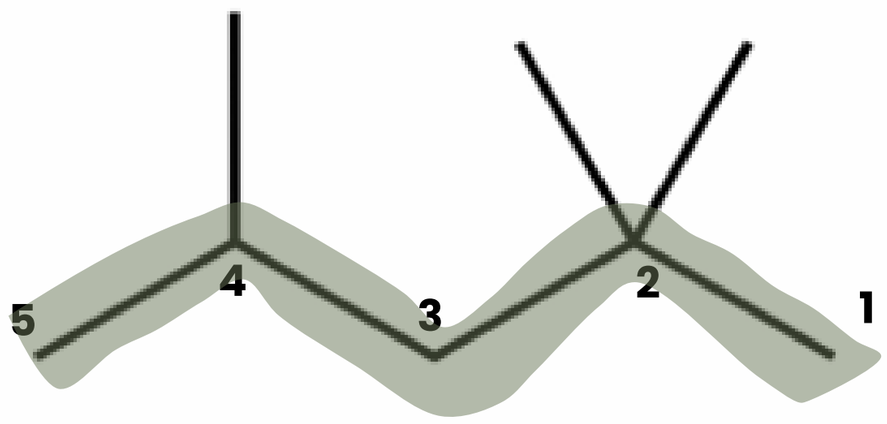
- ☑ LONGEST CHAIN CONFIRMED
- ☑ LOWEST LOCANTS
- ☑ ALPHABETICAL ORDER
- ☑ PUNCTUATION
- ☑ FORMULA COUNTED

A05 DEVELOPING

C_8H_{18}

PART A
STRUCTURE
> NAME

1. NUMBERED PARENT



2. NAMING TRAIL

PARENT: LONGEST CHAIN = 5 C → PENTANE.
NUMBERING: ONE DIRECTION GIVES 2,2,4; THE OTHER GIVES 2,4,4. FIRST POINT OF DIFFERENCE: 2 < 4, SO NUMBER FROM THE END THAT GIVES 2,2,4.
SUBSTITUENTS: METHYL ON C2, C2 AND C4 → 2,2,4-TRIMETHYL.

3. MY NAME

2,2,4-trimethylpentane

4. THE ALKANE CHECK



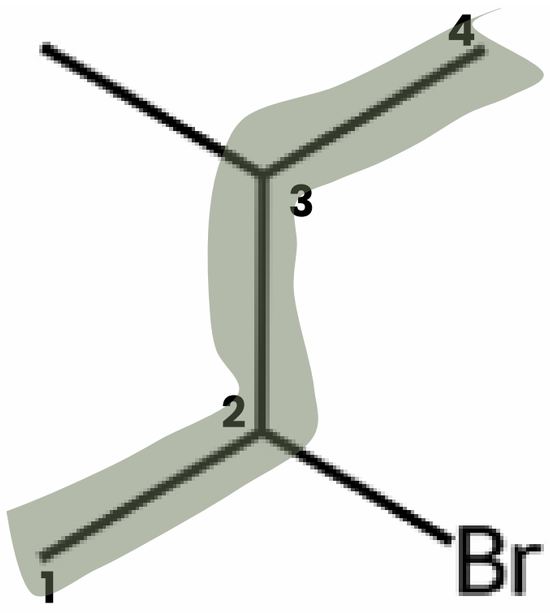
2,2,4-trimethylpentane

MATCH

- ☑ LONGEST CHAIN CONFIRMED
- ☑ LOWEST LOCANTS
- ☑ ALPHABETICAL ORDER
- ☑ PUNCTUATION
- ☑ FORMULA COUNTED

A06 DEVELOPING $C_5H_{11}BR$ PART A STRUCTURE > NAME

1. NUMBERED PARENT



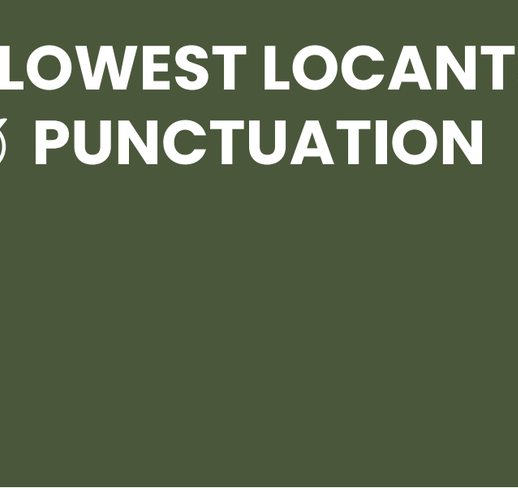
4. NAMING TRAIL

PARENT: LONGEST CHAIN = 4 C → BUTANE.
NUMBERING: BOTH DIRECTIONS GIVE THE SET {2,3}, SO IT IS A TIE.
ALPHABETICAL TIEBREAK: BROMO COMES BEFORE METHYL, SO BROMO GETS C2.
SUBSTITUENTS: BROMO ON C2, METHYL ON C3 → 2-BROMO-3-METHYL.

3. MY NAME

2-bromo-3-methylbutane

5. THE ALKANE CHECK



2-bromo-3-methylbutane

MATCH

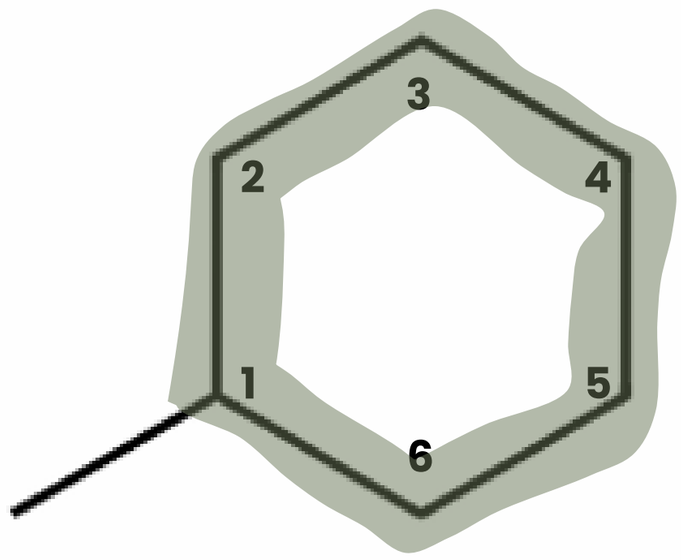
- ☑ LONGEST CHAIN CONFIRMED
- ☑ ALPHABETICAL ORDER
- ☑ FORMULA COUNTED
- ☑ LOWEST LOCANTS
- ☑ PUNCTUATION

A07 DEVELOPING

C_7H_{14}

PART A
STRUCTURE
> NAME

1. NUMBERED PARENT



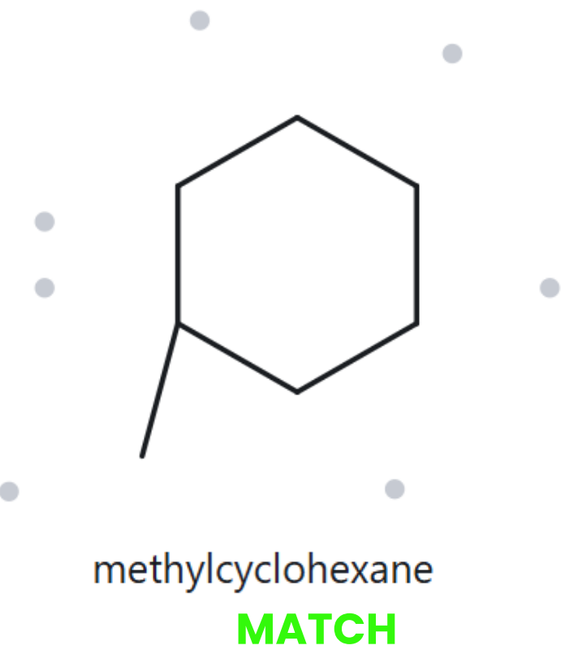
4. NAMING TRAIL

PARENT: A RING IS PRESENT, SO THE RING IS THE PARENT: 6 C → CYCLOHEXANE.
NUMBERING: THERE IS ONLY ONE SUBSTITUENT, SO NO LOCANT IS NEEDED.
SUBSTITUENTS: METHYL → METHYLCYCLOHEXANE.

3. MY NAME

methylcyclohexane

5. THE ALKANE CHECK



methylcyclohexane
MATCH

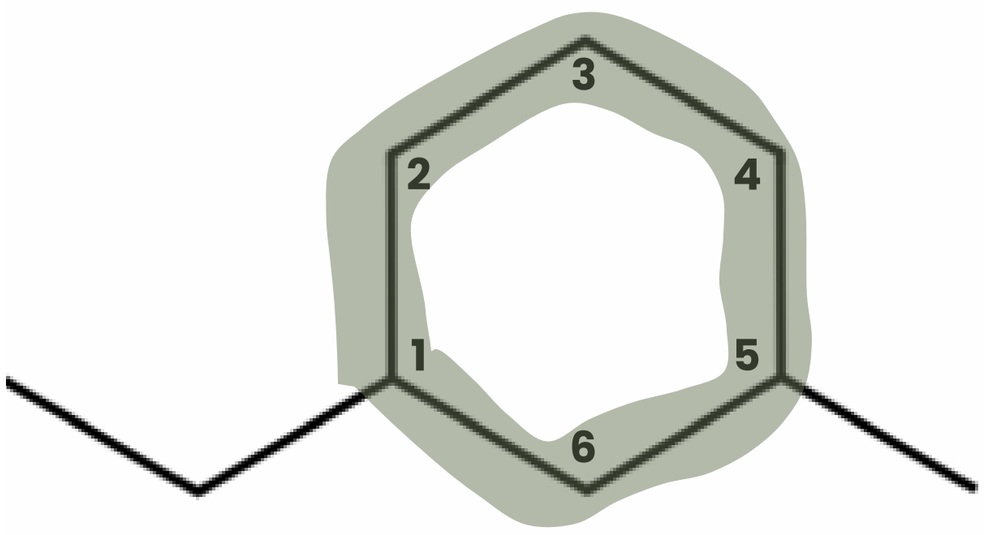
- ☑ LONGEST CHAIN CONFIRMED
- ☑ ALPHABETICAL ORDER
- ☑ FORMULA COUNTED
- ☑ LOWEST LOCANTS
- ☑ PUNCTUATION

A08 CHALLENGE

C_9H_{18}

PART A
STRUCTURE
> NAME

1. NUMBERED PARENT



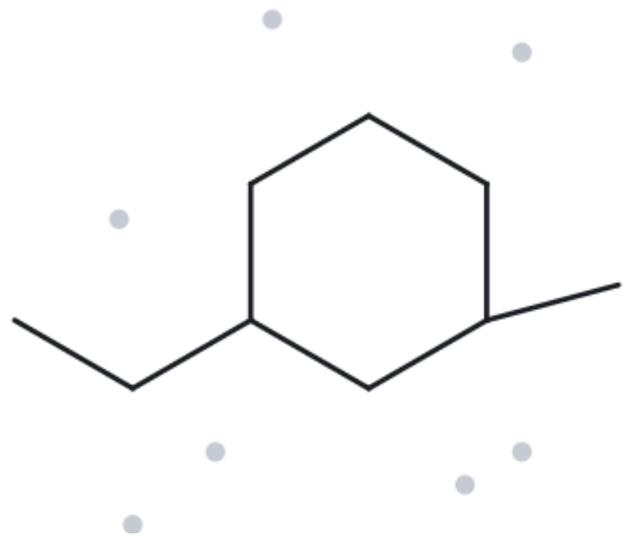
4. NAMING TRAIL

PARENT: RING OF 6 C → CYCLOHEXANE.
NUMBERING: THE LOCANT SET IS {1,3} IN BOTH DIRECTIONS AROUND THE RING, SO IT IS A TIE. ALPHABETICAL TIEBREAK: ETHYL COMES BEFORE METHYL, SO ETHYL GETS C1.
SUBSTITUENTS: ETHYL ON C1, METHYL ON C3 → 1-ETHYL-3-METHYL.

3. MY NAME

1-ethyl-3-methylcyclohexane

5. THE ALKANE CHECK



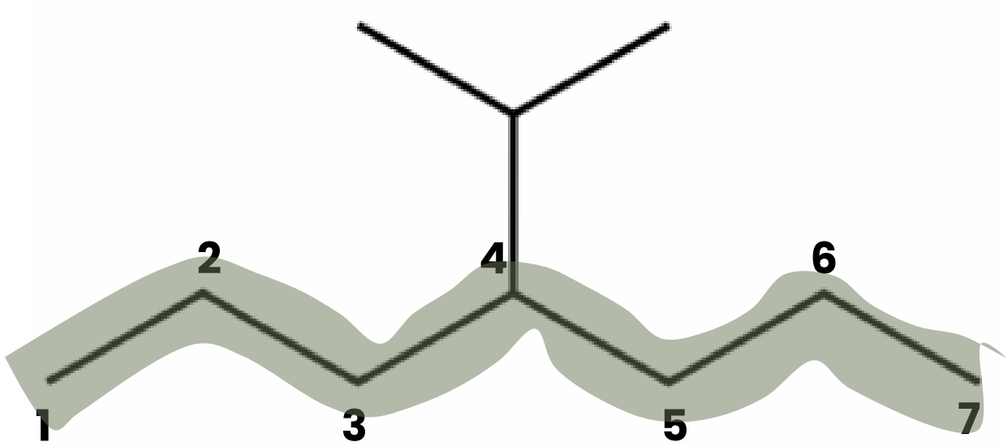
1-ethyl-3-methylcyclohexane
MATCH

- ☑ LONGEST CHAIN CONFIRMED
- ☑ LOWEST LOCANTS
- ☑ ALPHABETICAL ORDER
- ☑ PUNCTUATION
- ☑ FORMULA COUNTED

A09 CHALLENGE

$C_{10}H_{22}$ PART A STRUCTURE > NAME

1. NUMBERED PARENT



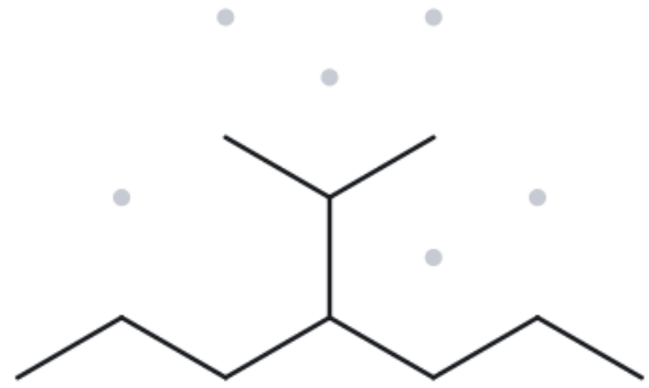
4. NAMING TRAIL

PARENT: LONGEST CHAIN = 7 C → HEPTANE. THE PATH THROUGH THE BRANCH IS ONLY 6 C, SO IT IS NOT THE PARENT.
NUMBERING: THE BRANCH IS ON C4 FROM EITHER END, SO THE NUMBERING IS THE SAME IN BOTH DIRECTIONS.
SUBSTITUENTS: 1-METHYLETHYL (ISOPROPYL) ON C4 → 4-(1-METHYLETHYL).

3. MY NAME

4-isopropylheptane

5. THE ALKANE CHECK



2-methyl-3-propylhexane

MISMATCH

BOTH NAMES DESCRIBE THE SAME COMPOUND ($C_{10}H_{22}$), BUT THEY CHOOSE DIFFERENT PARENTS. THE LONGEST CONTINUOUS CHAIN IS 7 C (HEPTANE), RUNNING FROM ONE PROPYL ARM THROUGH THE BRANCH CARBON INTO THE OTHER PROPYL ARM. THE ALKANE USED A 6 C PARENT (HEXANE) AND TREATED PART OF THE 7 C CHAIN AS A PROPYL BRANCH.

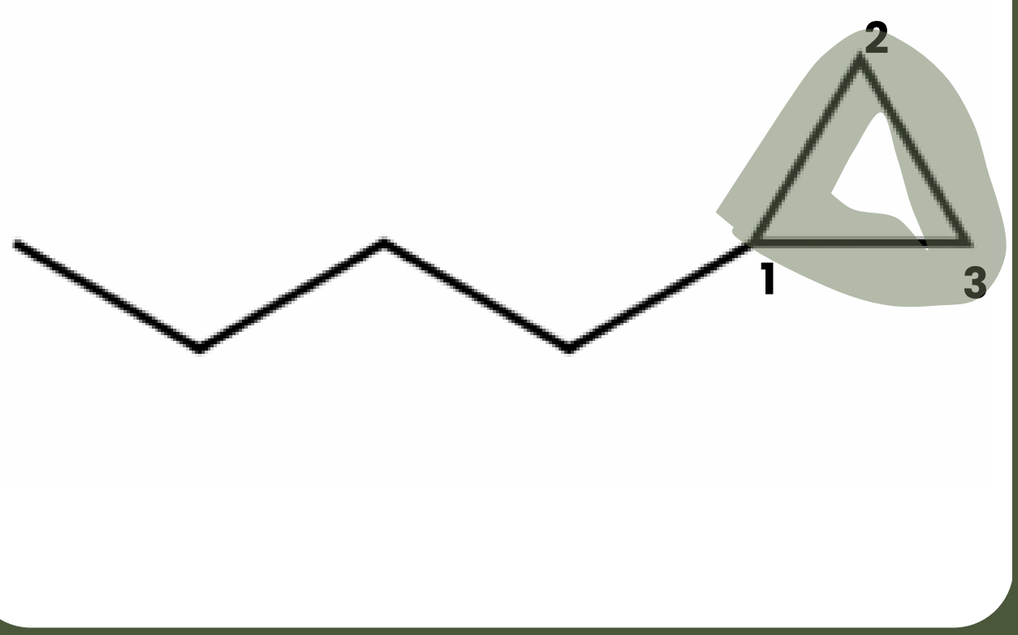
- ☑ LONGEST CHAIN CONFIRMED
- ☑ LOWEST LOCANTS
- ☑ ALPHABETICAL ORDER
- ☑ PUNCTUATION
- ☑ FORMULA COUNTED

A10 CHALLENGE

C_7H_{14}

PART A
STRUCTURE
> NAME

1. NUMBERED PARENT



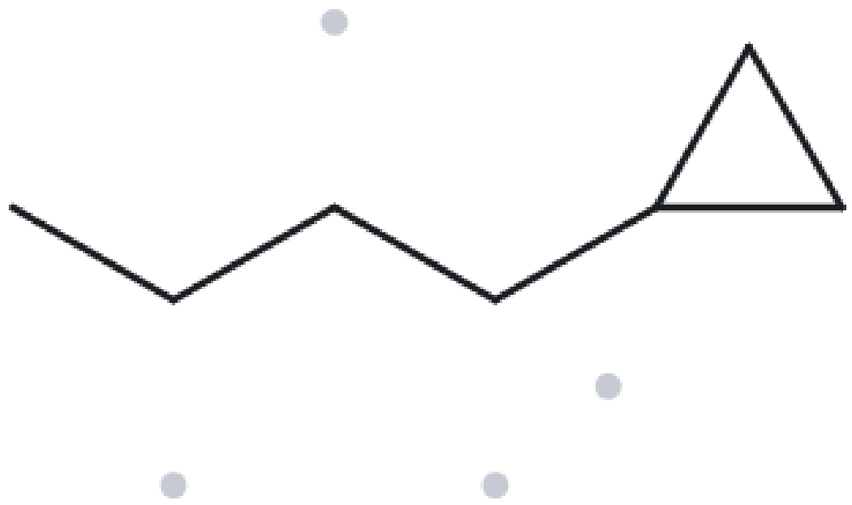
4. NAMING TRAIL

PARENT: A RING IS PRESENT, SO THE RING IS THE PARENT: 3 C → CYCLOPROPANE. THE 4-C CHAIN IS A SUBSTITUENT.
NUMBERING: THERE IS ONLY ONE SUBSTITUENT, SO NO LOCANT IS NEEDED.
SUBSTITUENTS: BUTYL → BUTYLCYCLOPROPANE.

3. MY NAME

butylcyclopropane

5. THE ALKANE CHECK



butylcyclopropane

MATCH

- ☑ LONGEST CHAIN CONFIRMED
- ☑ LOWEST LOCANTS
- ☑ ALPHABETICAL ORDER
- ☑ PUNCTUATION
- ☑ FORMULA COUNTED

PART B: STRUCTURE > NAME

B01 FOUNDATION

2-methylpropane

1. NAME DECODED

PARENT: PROPANE, A 3-CARBON CHAIN.

SUBSTITUENT: METHYL ON C2.

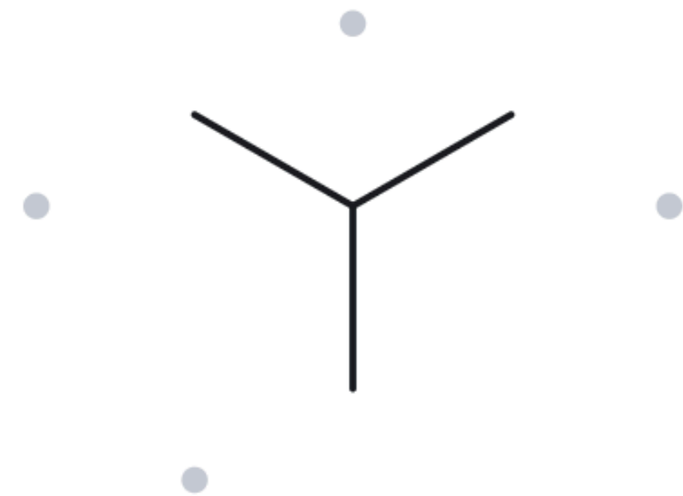
EXPECTED FORMULA: 3 + 1 = 4 C; OPEN CHAIN → C_4H_{10}.

3. FORMULA CHECK

COUNTED FROM THE DRAWING: C_4H_{10}

OPEN CHAIN: C_nH_{2n+2} = C_4H_{10} ✓

4. BOND-LINE STRUCTURE (THE ALKANE)



5. BACK CHECK

RETURNED BY THE ALKANE:
2-methylpropane

SAME AS THE GIVEN NAME

- ☑ PARENT LENGTH MATCHES
- ☑ FORMULA AGREES
- ☑ SUBSTITUENT POSITIONS
- ☑ BACK-CHECK RECORDED

PART B: STRUCTURE > NAME

B02 FOUNDATION

3-ethylpentane

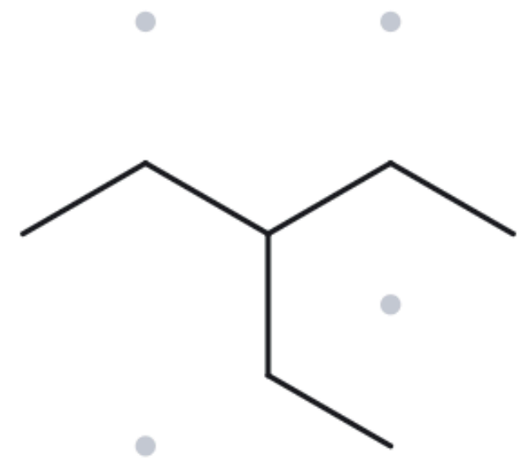
1. NAME DECODED

PARENT: PENTANE, A 5-CARBON CHAIN.

SUBSTITUENT: ETHYL ON C3.

EXPECTED FORMULA: 5 + 2 = 7 C; OPEN CHAIN → C_7H_{16}.

4. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

COUNTED FROM THE DRAWING: C_7H_{16}

OPEN CHAIN: C_nH_{2n+2} = C_7H_{16} ✓

5. BACK CHECK

RETURNED BY THE ALKANE:
3-ethylpentane

SAME AS THE GIVEN NAME

- ☑ PARENT LENGTH MATCHES
- ☑ FORMULA AGREES
- ☑ SUBSTITUENT POSITIONS
- ☑ BACK-CHECK RECORDED

PART B: STRUCTURE > NAME

B03 FOUNDATION

2,2-dimethylbutane

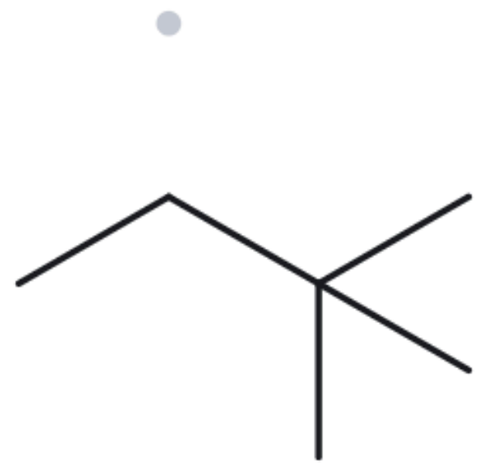
1. NAME DECODED

PARENT: BUTANE, A 4-CARBON CHAIN.

SUBSTITUENTS: TWO METHYL GROUPS, BOTH ON C2.

EXPECTED FORMULA: 4 + 1 + 1 = 6 C; OPEN CHAIN → C_6H_{14}.

4. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

COUNTED FROM THE DRAWING: C_6H_{14}

OPEN CHAIN: C_nH_{2n+2} = C_6H_{14} ✓

5. BACK CHECK

RETURNED BY THE ALKANE:
2,2-dimethylbutane

SAME AS THE GIVEN NAME

- ☑ PARENT LENGTH MATCHES
- ☑ FORMULA AGREES
- ☑ SUBSTITUENT POSITIONS
- ☑ BACK-CHECK RECORDED

PART B: STRUCTURE > NAME

B04 DEVELOPING

3-ethyl-2-methylpentane

1. NAME DECODED

PARENT: PENTANE, A 5-CARBON CHAIN.

SUBSTITUENTS: METHYL ON C2; ETHYL ON C3.

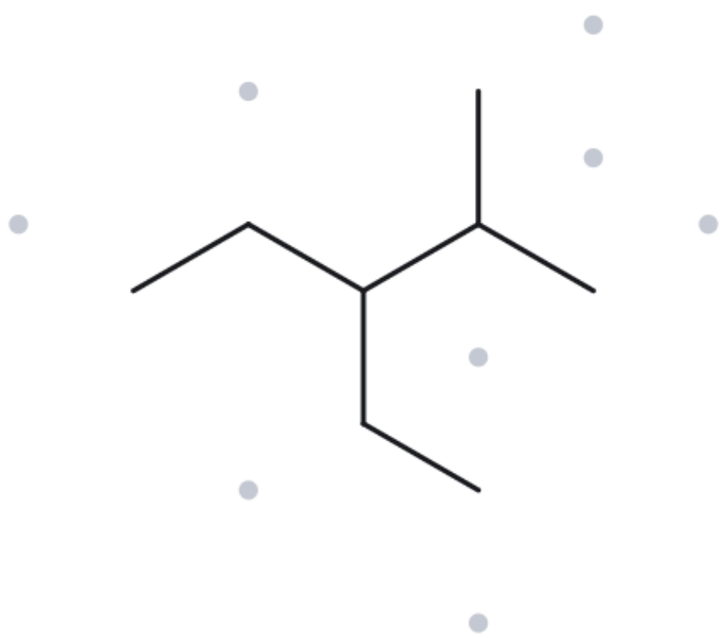
EXPECTED FORMULA: 5 + 1 + 2 = 8 C; OPEN CHAIN → C_8H_{18}.

3. FORMULA CHECK

COUNTED FROM THE DRAWING: C_8H_{18}

OPEN CHAIN: C_nH_{2n+2} = C_8H_{18} ✓

4. BOND-LINE STRUCTURE (THE ALKANE)



5. BACK CHECK

RETURNED BY THE ALKANE:
3-ethyl-2-methylpentane

SAME AS THE GIVEN NAME

- ☑ PARENT LENGTH MATCHES
- ☑ FORMULA AGREES
- ☑ SUBSTITUENT POSITIONS
- ☑ BACK-CHECK RECORDED

PART B: STRUCTURE > NAME

B05 DEVELOPING

1-chloro-2-methylpropane

1. NAME DECODED

PARENT: PROPANE, A 3-CARBON CHAIN.

SUBSTITUENTS: CHLORO ON C1; METHYL ON C2.

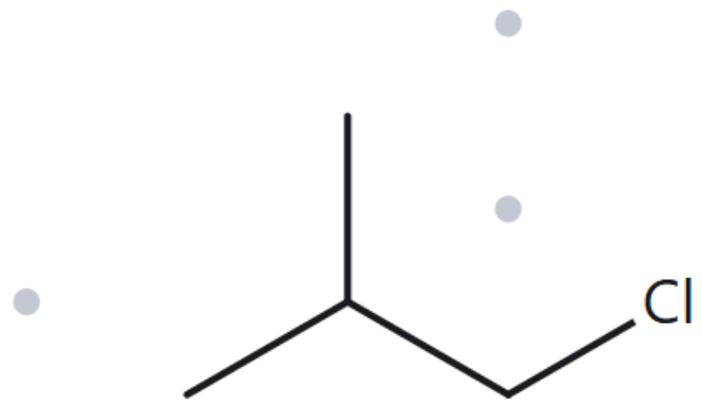
EXPECTED FORMULA: 3 + 1 = 4 C; OPEN CHAIN → C_4H_{10}; CL REPLACES ONE H → C_4H_9CL.

3. FORMULA CHECK

Counted from the drawing: C_4H_9Cl

Open chain: C_nH_{2n+2} = C_4H_{10}; minus one H for Cl = C_4H_9Cl ✓

4. BOND-LINE STRUCTURE (THE ALKANE)



5. BACK CHECK

RETURNED BY THE ALKANE:
1-chloro-2-methylpropane

SAME AS THE GIVEN NAME

- ☑ PARENT LENGTH MATCHES
- ☑ FORMULA AGREES
- ☑ SUBSTITUENT POSITIONS
- ☑ BACK-CHECK RECORDED

PART B: STRUCTURE > NAME

B06 DEVELOPING

1,2-dimethylcyclopentane

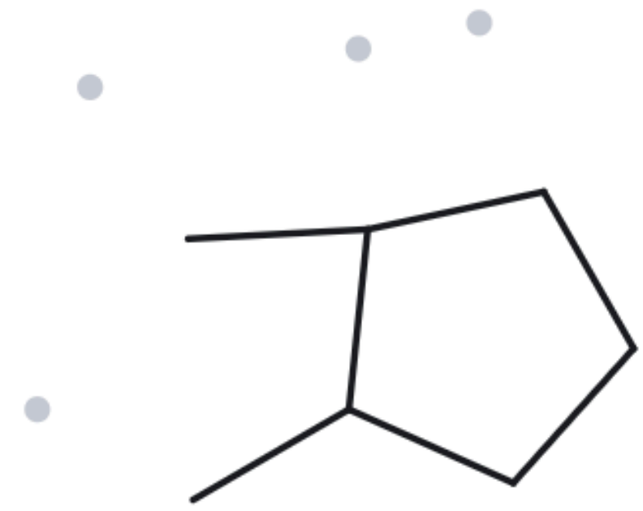
1. NAME DECODED

PARENT: CYCLOPENTANE, A 5-CARBON RING.

SUBSTITUENTS: METHYL ON C1; METHYL ON C2.

EXPECTED FORMULA: 5 + 1 + 1 = 7 C; ONE RING → C_7H_{14}.

4. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_7H_{14}

One ring: C_nH_{2n} = C_7H_{14} ✓

5. BACK CHECK

RETURNED BY THE ALKANE:
1,2-dimethylcyclopentane

SAME AS THE GIVEN NAME

- ☑ PARENT LENGTH MATCHES
- ☑ FORMULA AGREES
- ☑ SUBSTITUENT POSITIONS
- ☑ BACK-CHECK RECORDED

PART B: STRUCTURE > NAME

B07 DEVELOPING

4-ethyl-3,3-dimethylheptane

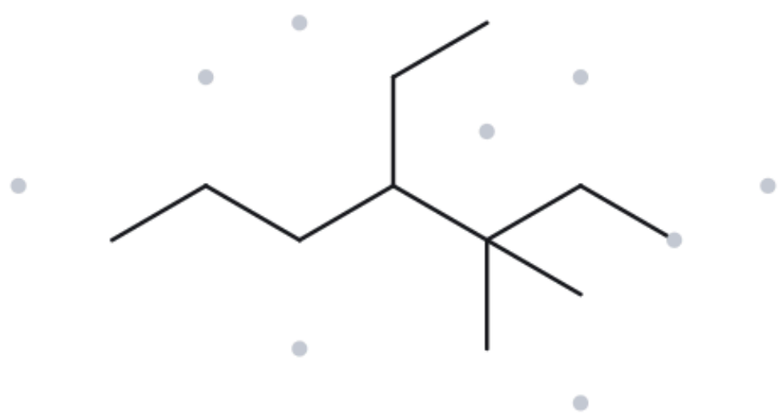
1. NAME DECODED

PARENT: HEPTANE, A 7-CARBON CHAIN.

SUBSTITUENTS: TWO METHYL GROUPS ON C3; ETHYL ON C4.

EXPECTED FORMULA: 7 + 1 + 1 + 2 = 11 C; OPEN CHAIN → $C_{11}H_{24}$.

4. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{11}H_{24}$

Open chain: C_nH_{2n+2} = $C_{11}H_{24}$ ✓

5. BACK CHECK

RETURNED BY THE ALKANE:
4-ethyl-3,3-dimethylheptane

SAME AS THE GIVEN NAME

- ☑ PARENT LENGTH MATCHES
- ☑ FORMULA AGREES
- ☑ SUBSTITUENT POSITIONS
- ☑ BACK-CHECK RECORDED

PART B: STRUCTURE > NAME

B08 CHALLENGE

tert-butylcyclohexane

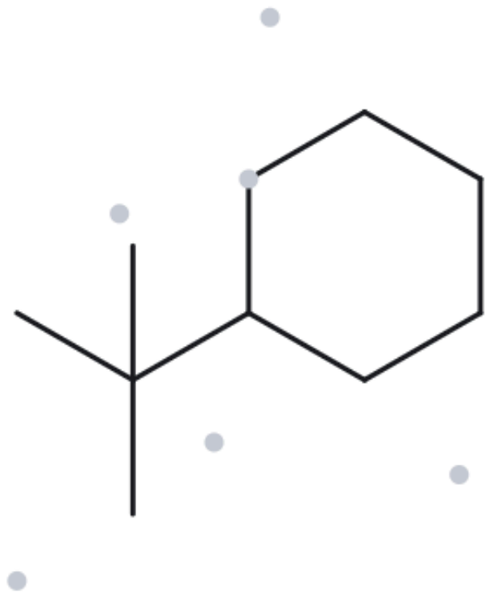
1. NAME DECODED

PARENT: CYCLOHEXANE, A 6-CARBON RING.

SUBSTITUENT: TERT-BUTYL, $-C(CH_3)_3$, ON C1 (NO LOCANT NEEDED FOR A SINGLE SUBSTITUENT).

EXPECTED FORMULA: 6 + 4 = 10 C; ONE RING → $C_{10}H_{20}$.

4. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

One ring: C_nH_{2n} = $C_{10}H_{20}$ ✓

5. BACK CHECK

RETURNED BY THE ALKANE:
(1,1-dimethylethyl)cyclohexane

NOT THE SAME AS THE GIVEN NAME

THE ALKANE USES THE SYSTEMATIC NAME FOR THE SUBSTITUENT, WHILE THE GIVEN NAME USES THE COMMON NAME. TERT-BUTYL, $-C(CH_3)_3$, IS NAMED SYSTEMATICALLY AS (1,1-DIMETHYLETHYL): AN ETHYL GROUP WITH TWO METHYLS ON C1, THE CARBON ATTACHED TO THE RING.

- ☑ PARENT LENGTH MATCHES
- ☑ FORMULA AGREES
- ☑ SUBSTITUENT POSITIONS
- ☑ BACK-CHECK RECORDED

PART B: STRUCTURE > NAME

B09 CHALLENGE

3-ethyl-1,1-dimethylcyclohexane

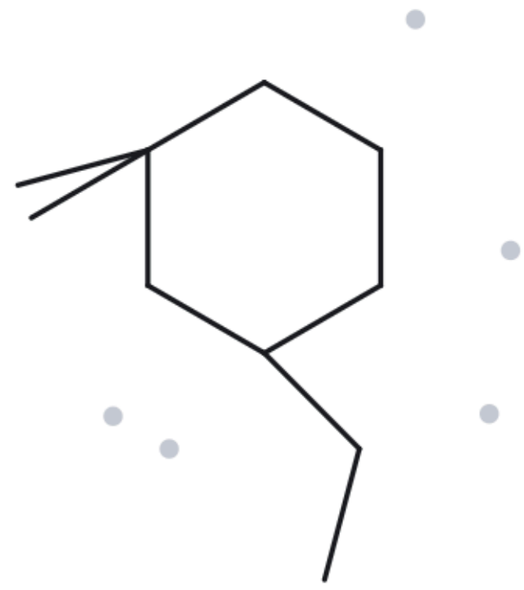
1. NAME DECODED

PARENT: CYCLOHEXANE, A 6-CARBON RING.

SUBSTITUENTS: TWO METHYL GROUPS ON C1; ETHYL ON C3.

EXPECTED FORMULA: 6 + 1 + 1 + 2 = 10 C; ONE RING → $C_{10}H_{20}$.

4. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

One ring: $C_nH_{2n} = C_{10}H_{20}$ ✓

5. BACK CHECK

RETURNED BY THE ALKANE:
3-ethyl-1,1-dimethylcyclohexane

NOT THE SAME AS THE GIVEN NAME

THE GIVEN NAME BREAKS THE LONGEST-CHAIN RULE. IN 2-ETHYLBUTANE, THE "ETHYL" BRANCH ON C2 IS NOT A TRUE BRANCH: IT CONTINUES THE CHAIN. TRACING FROM ONE END THROUGH C2 AND INTO THE ETHYL GROUP GIVES 5 CARBONS ($CH_3CH_2–CH–CH_2CH_3$), NOT 4. THE PARENT MUST BE PENTANE, NOT BUTANE.

- ☑ PARENT LENGTH MATCHES
- ☑ FORMULA AGREES
- ☑ SUBSTITUENT POSITIONS
- ☑ BACK-CHECK RECORDED

PART B: STRUCTURE > NAME

B10 CHALLENGE

3-ethyl-1,1-dimethylcyclohexane

1. NAME DECODED

PARENT: BUTANE, A 4-CARBON CHAIN.

SUBSTITUENT: ETHYL ON C2.

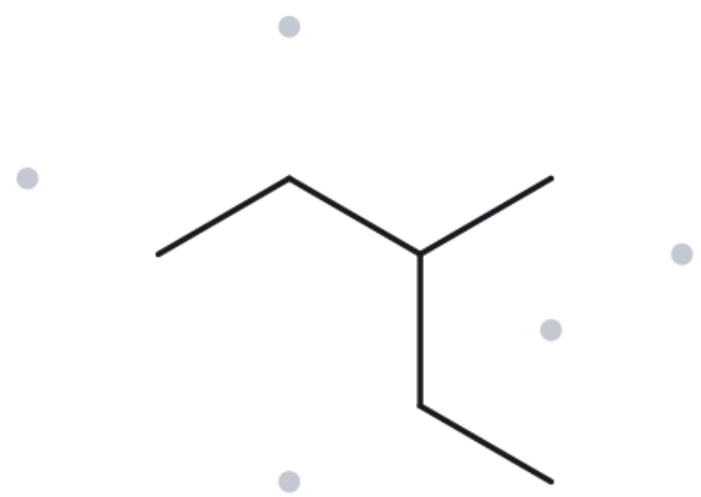
EXPECTED FORMULA: 4 + 2 = 6 C; OPEN CHAIN → C_6H_{14}.

3. FORMULA CHECK

Counted from the drawing: C_6H_{14}

Open chain: C_nH_{2n+2} = C_6H_{14} ✓

4. BOND-LINE STRUCTURE (THE ALKANE)



5. BACK CHECK

RETURNED BY THE ALKANE:
3-methylpentane

SAME AS THE GIVEN NAME

- ☑ PARENT LENGTH MATCHES
- ☑ FORMULA AGREES
- ☑ SUBSTITUENT POSITIONS
- ☑ BACK-CHECK RECORDED